

TAMIL NADU STATE BOARD - STANDARD 12 CHEMISTRY VOLUME 1

CHAPTER 7: CHEMICAL KINETICS

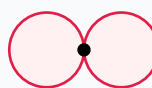
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 12 English Medium
Subject:	Chemistry Volume 1
Chapter Name:	Chemical Kinetics

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)



p orbital (px)

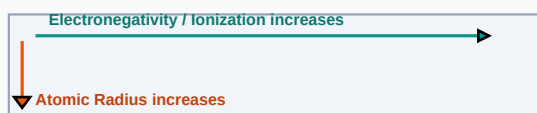
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Rate of Reaction	The change in concentration of a reactant or product per unit time in a chemical process.
Order of Reaction	The sum of the concentration exponents of reactants in the experimentally derived rate law equation.
Half-Life	The time required for reactant concentration to decrease to exactly half of its initial value.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

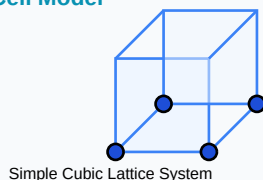
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Essential Formulas, Laws & Identities:

- First Order Rate: $k = (2.303 / t) * \log_{10}([A]_0 / [A])$
- First Order Half-Life: $t_{0.5} = 0.693 / k$
- Arrhenius Equation: $k = A * e^{(-E_a / R * T)}$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

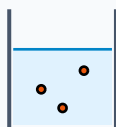
Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: A first-order reaction has $k = 0.0693 \text{ min}^{-1}$. Find its half-life.

Step-by-step Solution:

$$t_{0.5} = 0.693 / k = 0.693 / 0.0693 = 10 \text{ minutes.}$$

Titration Beaker & Solute Precipitation



Reacting Mixture

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Derive the integrated rate equation for a first-order chemical reaction.

Analytical Solution Strategy:

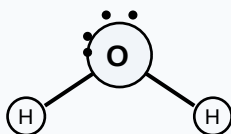
Rate = $-d[A]/dt = k * [A] \Rightarrow d[A]/[A] = -k dt$.

Integrate: $\ln([A]) = -kt + C$. At $t=0$, $[A]=[A]_0 \Rightarrow C = \ln([A]_0)$.

$\ln([A]) = -kt + \ln([A]_0) \Rightarrow kt = \ln([A]_0 / [A])$.

Convert to base 10: $k = (2.303 / t) * \log_{10}([A]_0 / [A])$.

Lewis Dot Structure (H₂O Bent Molecular Geometry)



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

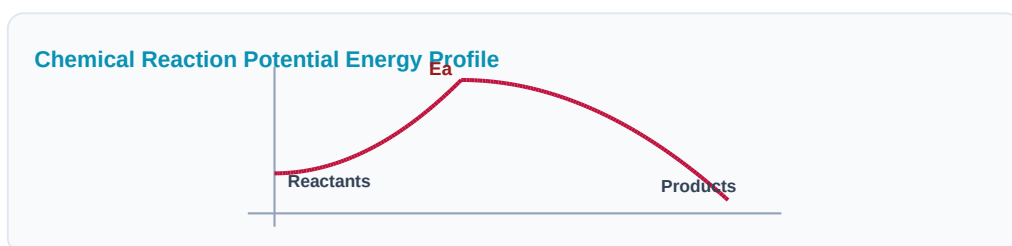
Q1 (1-Mark): Differentiate between rate of reaction and rate constant.

Q2 (2-Mark): Explain molecularity of a reaction. How does it differ from order?

Q3 (2-Mark): State the Arrhenius equation and write down its logarithmic form.

Q4 (5-Mark): Show that half-life of zero-order reaction is proportional to initial concentration.

Q5 (5-Mark): Explain collision theory and activation energy.



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Pharmacology

Drug half-lives are evaluated to calibrate dosing intervals, keeping bloodstream levels safe.

Application Case: Archaeology

Carbon-14 half-life of 5730 years calibrates dating timelines of fossil relics.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C₆H₆) Resonance



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Electrochemical Galvanic Cell System

